



CONCEPT CHECKLIST
DCIT60 – METHODS OF RESEARCH

Title: **FLOODSENSE: AN ANDROID APPLICATION USING AN EXPERT SYSTEM FOR FLOOD SUSCEPTIBILITY ASSESSMENT AND PRE-EVENT PREPAREDNESS IN BACOR CITY**

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SCOPE	LIMITATIONS
FloodSense is limited geographically to Bacoor City, Cavite. The system will present flood-susceptibility information only for barangays or smaller approved geographic zones within Bacoor City for which sufficient spatial data, historical flood evidence, and expert validation are available. The final geographic resolution - barangay, sub-barangay, or expert-defined susceptibility zone - will follow the most precise level that the approved data can reliably support.	The system does not cover municipalities or cities outside Bacoor City and will not extend conclusions to adjacent Local Government Units (LGUs) such as Imus, Kawit, or Las Pinas. It will not claim street-, household-, or property-level precision unless the approved evidence explicitly supports that resolution. Areas without enough approved evidence may return an 'Insufficient Data' result rather than an assumed susceptibility level.
The system is scoped to pre-event, rainfall-induced flood-susceptibility assessment. Users will select a Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA)-aligned rainfall intensity category and a rainfall duration such as 1, 3, 6, 12, or 24 hours. Exact numerical thresholds used internally will follow the verified PAGASA reference adopted by the study before final deployment.	FloodSense is not a real-time weather service and does not automatically retrieve or interpret live rainfall, radar, river-level, drainage, or sensor feeds in the core system. The user-selected rainfall scenario is a planning input, not proof of current conditions. Tidal flooding, storm surge, tsunami, dam-break flooding, and other hazards outside the approved rainfall-induced scope are not classified unless separately added and validated in a future version.
The knowledge base of the Expert System will be constructed from data and domain knowledge supplied or validated by authorized subject-matter experts and institutional sources, principally the Bacoor Disaster Risk Reduction and Management Office (BDRRMO), with supporting spatial or planning information from the Bacoor City Planning and Development Office (CPDCO) and rainfall references from PAGASA or other approved providers. Each public-facing rule will be linked to its source, review status, version, and approval record.	The quality of the Expert System is bounded by the completeness, geographic coverage, recency, and reliability of the information that authorized institutions are able to release and validate. Data access, final expert reviewers, and some datasets may remain pending during development. Provisional demonstration data may be used for coding and testing only when clearly labelled and must not be presented as an official or validated public result.



FloodSense will use a Rule-Based Expert System. Expert-provided knowledge will be encoded into structured facts, decision tables, and approved IF-THEN rules. A forward-chaining inference process will evaluate the user's selected location, rainfall intensity, rainfall duration, applicable geographic profile, and other approved conditions, then identify the rule or rules that apply. The system will return an explainable susceptibility conclusion instead of relying on an opaque prediction model.

The Expert System is not self-learning and will not automatically create new rules from user behavior or unverified internet information. Machine Learning (ML) classification is no longer part of the revised system. The rule base must be reviewed and updated when authoritative data or expert guidance changes. The system cannot account for unrecorded or real-time conditions such as sudden drainage blockage, localized construction, soil saturation, or rapidly changing water levels unless those factors are later supplied as approved inputs.

The Expert System will classify supported geographic areas under the selected rainfall scenario using the approved output levels Low, Moderate, High, and Very High. Where equally applicable approved rules disagree, the system may return 'Uncertain'; where no approved rule covers the scenario, it may return 'Insufficient Data.' Each result will include the selected scenario, geographic scope, applicable rule or explanation, source/version information, and a reminder that the result is a susceptibility estimate.

A susceptibility classification is not the same as a flood forecast, flood-risk probability, live flood status, evacuation order, or guarantee that flooding will or will not occur. FloodSense will not independently declare a road safe, predict exact flood depth or arrival time, or guarantee user safety. Users must continue to follow current instructions and warnings from BDRRMO, PAGASA, barangay authorities, and other official emergency agencies.

The Android application will contain a dynamic, movable Geographic Information System (GIS)-style map. Users can pan, zoom, tap supported areas, open information pop-ups, show approved map layers, and view evacuation-center markers. Susceptibility polygons will be color-coded according to the current Expert System result - Low (green), Moderate (yellow), High (orange), and Very High (red). When rainfall intensity or duration changes, the same stored geographic polygons can be re-evaluated and recolored without creating a new static map image.

The dynamic map visualizes approved spatial data and Expert System outputs; it is not a live inundation map. Map precision is limited by the quality and scale of the approved boundaries and susceptibility zones. Base-map availability, map-tile detail, and external mapping services may depend on network connectivity and provider terms. The mobile user may inspect and select map features but cannot edit official polygons, evacuation-center records, or susceptibility rules.

A separate Web Administration System will provide authorized administrators with map-maintenance tools. Using a web map, an administrator may add, move, edit, deactivate, or remove approved point features such as evacuation centers and may draw or edit approved polygon features such as barangay boundaries or susceptibility zones. Geographic edits will be stored as spatial data in PostgreSQL with the PostGIS extension and processed

Map editing is restricted to authorized administrative roles and does not give an administrator permission to arbitrarily assign flood susceptibility by manually painting areas red, orange, yellow, or green. Public susceptibility colors must come from the active Expert System result. Geographic edits and other critical changes should be reviewed, versioned, and published through the approved workflow before they are exposed to



through GeoDjango. Administrators may also manage labels, metadata, source references, and publication status for map records.	mobile users. Incorrect edits remain possible if administrative validation procedures are not followed.
The 'Check My Area' feature will allow the user to manually position or drag a pin on the dynamic Bacoor City map. The selected coordinates will be sent through the secured Application Programming Interface (API), and GeoDjango/PostGIS will resolve the point to the most specific approved geographic zone. The Expert System will then evaluate that zone using the active rainfall scenario and return the susceptibility level and explanation for the selected area.	Check My Area does not require automatic Global Positioning System (GPS) access in the core design and therefore does not automatically prove the user's actual physical position. Accuracy is limited by the user's pin placement and the approved boundary geometry. A pin placed near a boundary may resolve to a neighboring zone, so the interface must display the resolved area and allow the user to reposition the pin when necessary.
After the Expert System produces a susceptibility result, the Decision Support System (DSS) will provide pre-event preparedness support based on that result and on the user's supported question, selected concern, or answers to situational prompts. The DSS may guide the user about go-bag preparation, family communication, protection of documents and valuables, monitoring of official advisories, preparedness for vulnerable household members, evacuation readiness, and relevant evacuation resources. Its decision paths and recommendation content will be based on expert-approved preparedness knowledge.	The DSS supports decision-making but does not replace the judgment or legal authority of BDRRMO, PAGASA, barangay officials, emergency responders, or the user. It will not issue an independent mandatory evacuation order, guarantee that a recommended action is sufficient for every household, or answer unsupported real-time questions as if verified. An open-ended Artificial Intelligence (AI) assistant is not required for the core Minimum Viable Product (MVP) and should remain optional unless separately approved and validated.
The Web Administration System will include management functions for the DSS. Authorized administrators may create or revise approved situational questions, answer choices, decision branches, susceptibility-level conditions, preparedness recommendations, source references, activation status, and versions. This makes it possible to update approved preparedness content without changing and rebuilding the Flutter mobile application's source code for every content revision.	Only approved, active DSS content should be delivered to users. Administrative access to DSS editing must be role-restricted and logged. Changes that have not completed the required expert or institutional review must remain in draft or review status and must not immediately affect public guidance. The system does not automatically validate the factual correctness of content entered by an administrator.



The 'Find Nearest Evacuation Center' feature will display verified BDRRMO-designated or otherwise officially approved evacuation centers on the dynamic map and in a ranked list. The backend can use stored geographic coordinates to calculate straight-line proximity from the user's selected pin. Each center may display its name, barangay, address, verified status, last-validation date, and other fields that the responsible office officially provides. A 'Get Directions' action may open an external navigation application for route guidance.

Nearest-center ranking does not mean that the route is currently safe, passable, or the fastest route. FloodSense does not provide a custom routing engine, live traffic, road-closure detection, real-time flood-depth routing, or guaranteed travel time. Evacuation-center capacity, occupancy, opening status, and contact details are shown only when current verified information is available. External navigation accuracy and routing remain the responsibility of the selected mapping provider.

User registration and authentication will use an email address and password as the primary login credentials. Registration may additionally collect the user's full name or display name/username, home barangay, password confirmation, and acceptance of the privacy notice and FloodSense disclaimer. The system may support email-based password recovery and authenticated session management. User-facing authentication will be handled through the Django backend rather than Firebase Authentication.

Social-media login, phone-based One-Time Password (OTP) login, and biometric authentication are outside the core scope unless later approved. The system will collect only information required for approved functions and will not expose passwords to administrators. Account availability depends on access to the registered email for recovery functions. User accounts do not give users permission to edit Expert System rules, DSS content, official map geometry, or evacuation-center records.

The Web Administration System will be used by authorized administrators while residents use the Android application. Administrative functions will include managing Expert System knowledge and rule versions, DSS content, rainfall parameters, geographic zones and map features, evacuation-center records, data sources and provenance, user/account status where authorized, publication status, and audit history. The system will use role-based permissions so administrative actions are separated from ordinary user functions.

The Web Administration System is not a public resident portal. Administrative access must be restricted to authorized accounts, and sensitive operations must be logged. Administrators must not be able to view users' plain-text passwords. The system cannot guarantee correct decisions if unauthorized, inaccurate, or unreviewed data are deliberately entered; therefore, approval, versioning, backup, and audit procedures remain necessary operational controls.

The recommended technical architecture uses Flutter with Dart for the Android user application; Python Django for the backend and Web Administration System; Django REST Framework (DRF) for the secured mobile API; GeoDjango for geographic processing; PostgreSQL for relational data; PostGIS for spatial geometry and queries; flutter_map for the mobile interactive map; and Leaflet with map-editing tools for the administrator's web map. The Expert System and DSS

The initial user application is limited to Android; an iOS application is not included in the core thesis scope. The administrator requires a supported modern web browser. The revised system does not use Unity, Vuforia, Augmented Reality (AR), Firestore as the primary database, or Machine Learning (ML) as the susceptibility classifier. Additional infrastructure such as microservices, Kubernetes, Redis, or separate rule-engine platforms will not be added unless



logic will be implemented as maintainable Python/Django services using database-driven rules and content.	development later demonstrates a clear technical need.
PostgreSQL/PostGIS will serve as the central system database for approved geographic polygons and points, Expert System rules and versions, DSS structures, rainfall parameters, evacuation centers, source/provenance records, user and administrator records, and audit information. GeoDjango will perform spatial operations such as determining which approved zone contains a selected point and retrieving nearby verified evacuation centers. Critical records will include version, validation, and source metadata where applicable.	The database does not make unverified information authoritative by itself. Spatial accuracy, rule coverage, and output quality remain dependent on the source data and approval process. Direct database credentials will not be embedded in the Android application. Users and web clients will interact through controlled backend services. Full disaster-recovery guarantees, high-availability clustering, and enterprise-scale replication are outside the undergraduate MVP unless specifically required by the deployment environment.
FloodSense will apply data-minimization and security controls consistent with Republic Act No. 10173, or the Data Privacy Act of 2012. The backend will use authenticated and encrypted communication, role-based access for administrative functions, secure password storage, and audit logging for important administrative changes. A manually placed Check My Area pin is intended for the current assessment and does not need to be permanently stored in the user's profile unless a later approved requirement specifically justifies storage.	No system can guarantee absolute security. FloodSense will not intentionally collect unnecessary demographic, health, financial, or continuous-location data in the core design. If external map tiles or navigation services are opened, those services operate under their own privacy and service policies. Location accuracy, device security, network security, and user handling of account credentials are partly outside the control of FloodSense.
A first-launch onboarding and information module will explain the application's purpose, rainfall scenario controls, map legend, Expert System result meanings, DSS role, data date/version where applicable, Check My Area instructions, evacuation-resource behavior, and the safety disclaimer. Users will be able to revisit these instructions from the main dashboard. The interface will use plain language and clearly distinguish susceptibility information from official real-time warnings.	Onboarding and disclaimers cannot ensure that every user will fully understand or follow the information presented. FloodSense is an informational and decision-support application and does not guarantee safety, prevent flooding, or replace emergency services. The user remains responsible for following official warnings, evacuation directives, and emergency instructions from authorized government agencies.



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The project will prioritize a working core system consisting of user authentication, rainfall-scenario selection, dynamic susceptibility mapping, Check My Area, the Rule-Based Expert System, the Decision Support System, verified evacuation resources, the Django Web Administration System, and the supporting PostgreSQL/PostGIS database and secured API. Development will favor a practical architecture that the research team can build, test, explain, and maintain rather than adding unnecessary technologies.

Real-time forecasting, live flood sensors, live traffic or road-condition monitoring, real-time evacuation-center occupancy, crowdsourced flood reporting, three-dimensional flood simulation, Augmented Reality, Machine Learning classification, and a general-purpose AI chatbot are outside the core MVP unless separately approved as future enhancements. Internet connectivity is expected for fresh synchronization and server-side evaluation; complete offline operation is not guaranteed in the initial implementation.

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